# Robotic Budding: From Homogeneous Cells to Diverse Morphologies in Physical Self-Reproduction

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***Abstract*—Physical self-reproduction in modular robots remains difficult because a robot must not only move, but also collect, align, dock, and separate modules reliably in hardware. Analogous to homogeneous biological cells that self-organize into complex living tissues, we introduce a single standardized robotic module combining a magnetic docking interface with two actuated joints, that folds from a flat, two-dimensional configuration into three-dimensional structures. This single design can be assembled into two morphologies: a three-module "trilobite" and a five-module "seal". An assembled parent robot can be directed, via a small set of discrete motion commands, to locate and magnetically dock with a sequence of scattered spare modules, ultimately assembling them into a second, structurally identical individual that separates from the unchanged parent. This is a budding-style form of physical reproduction. Here, we report preliminary hardware demonstrations of this full process, establishing an end-to-end budding-style physical self-reproduction system across diverse morphologies. We discuss the current scope of the system, including its present reliance on direct human teleoperation rather than autonomous control, and outline directions toward autonomous execution of this process and repeated validation across a larger number of independent trials.**

## I. Introduction

Modular robotic systems achieve versatility, robustness, and adaptability by composing a small set of standardized units into task-specific morphologies [1]. A particularly ambitious extension of this idea is physical self-reproduction: using the same modules that make up a robot to construct additional, independently functioning copies of it. This goal's theoretical foundations date back to von Neumann's theory of self-reproducing automata [2]. Recent work has demonstrated kinematic self-replication in reconfigurable biological organisms [3], but achieving comparable physical reproduction in purely electromechanical systems has remained largely limited to simulation or to structured, partially scaffolded environments [4]–[6]. Existing hardware implementations have generally been limited to small module counts or to environments engineered to simplify the assembly process.

In this work, we investigate whether a single module design (Fig. 1) can support both (i) multiple, functionally distinct target morphologies, and (ii) physical reproduction via module collection and reassembly, without redesigning the hardware between cases. We report a system in which an assembled parent robot is directed, through a small set of discrete motion commands issued over a serial link to its onboard microcontroller, to locate and dock with scattered spare modules in sequence, culminating in the release of a structurally identical offspring.

To demonstrate reproduction across diverse body plans, we implement two structurally and topologically distinct representative morphologies as proof-of-concept: a three-module "trilobite-like" configuration featuring a segmented body with a short caudal extension, and a five-module "seal-like" configuration equipped with a pair of rear flippers, as shown in Fig. 2. This illustrates how module count and connection topology alone determine the resulting organismal diversity. We demonstrate the full collection-and-reproduction process on both morphologies.

The contribution of this extended abstract is a preliminary hardware demonstration and system description, rather than a comprehensive quantitative evaluation; a more complete study, including autonomous control and validation across both morphologies, is in preparation.

For a target morphology containing $n$ modules, let $\mathcal{O} = \{o_1, \dots, o_n\}$ denote the modules initially composing the parent and let $\mathcal{E} = \{e_1, \dots, e_n\}$ denote the modules acquired from the environment. The physical reproduction process considered here is therefore written as

$$R(\mathcal{O}) + \mathcal{E} \xrightarrow{\pi_{\mathrm{collect}}} R(\mathcal{O} \cup \mathcal{E}) \xrightarrow{\pi_{\mathrm{separate}}} R(\mathcal{O}) + R(\mathcal{E}), \quad (1)$$

where $\pi_{\mathrm{collect}}$ is the teleoperated module-acquisition action and $\pi_{\mathrm{separate}}$ is the pre-programmed separation sequence. Let $G_n^{\star}$ denote the target module-connection graph and let $G_{\mathrm{p}}$ and $G_{\mathrm{o}}$ denote the graphs of the separated parent and offspring. Structural reproduction requires

$$G_{\mathrm{p}} \simeq G_{\mathrm{o}} \simeq G_n^{\star}, \qquad |V(G_{\mathrm{p}})| = |V(G_{\mathrm{o}})| = n, \quad (2)$$

where $\simeq$ denotes graph isomorphism. To make the material lineage explicit, we define the module-provenance segregation index

$$\sigma = \frac{|V(G_{\mathrm{p}}) \cap \mathcal{O}| + |V(G_{\mathrm{o}}) \cap \mathcal{E}|}{2n}. \quad (3)$$

The demonstrated process has $\sigma = 1$: all original modules remain in the persistent parent, whereas all acquired modules constitute the offspring. Thus, although the transient $2n$-module composite separates into two equal-sized $n$-module robots, the reproduction mode is more accurately described as equal-sized budding-like reproduction rather than mitosis or binary fission.

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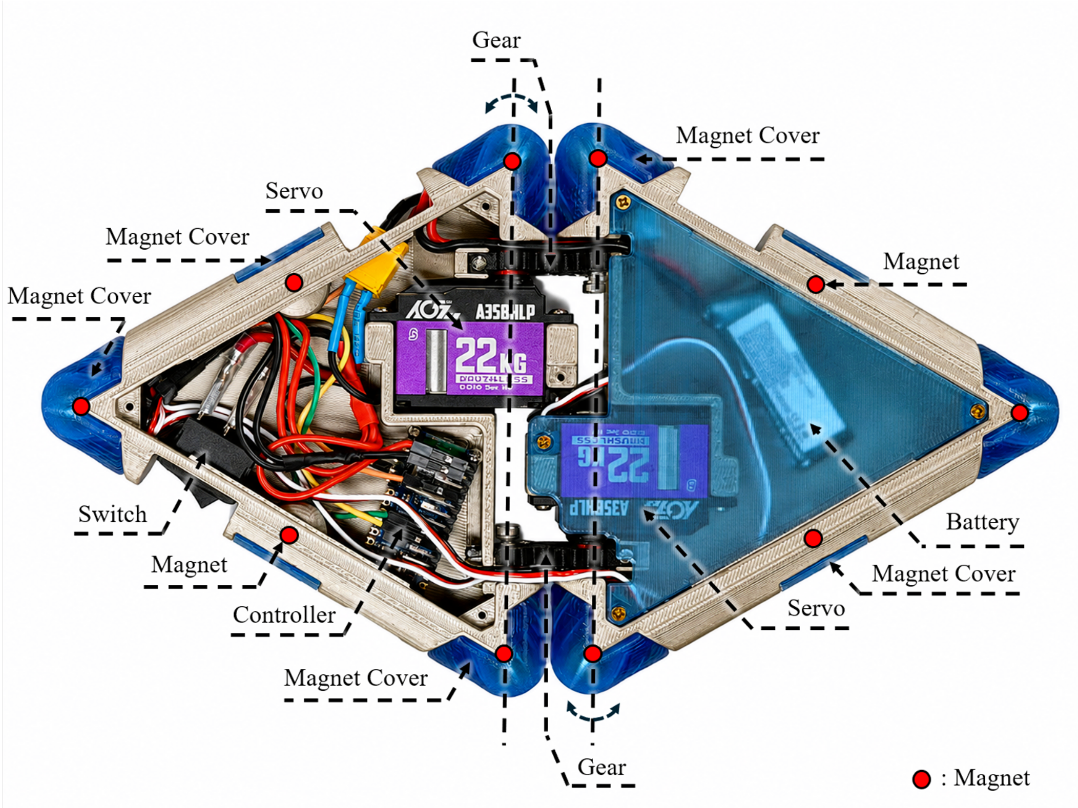



Fig. 1: Internal structural layout and component allocation of an individual module, highlighting the dual-servo drive mechanism, onboard electronics, and magnetic docking interfaces (red dots).

## II. System Overview

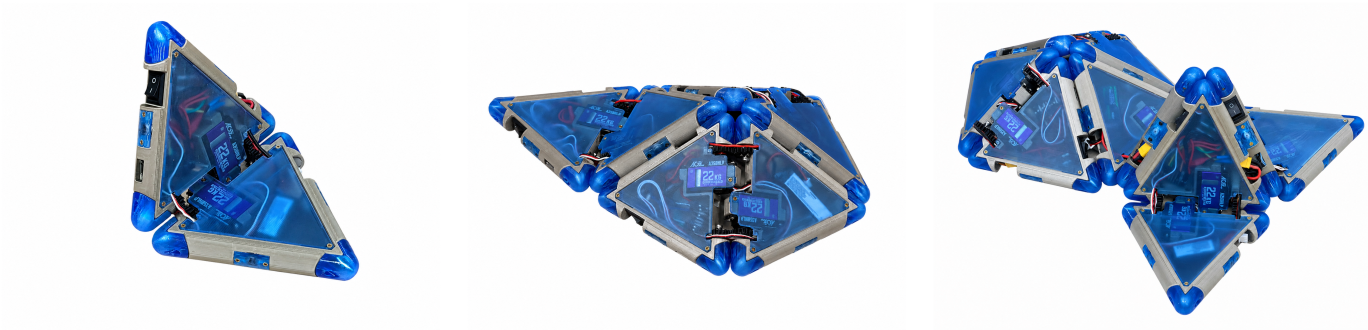
As shown in Fig. 1, each module is composed of two rigid triangular segments connected in series by two revolute joints. Each joint is driven by an internal servo actuator and position-controlled across approximately 90°, providing a total relative articulation range of $\pm 180°$ between the two segments. This range of motion allows the module to fold from a flat, planar configuration into three-dimensional structures. Internally, the hollow triangular frames house a 7.4 V Li-Po battery power unit and an onboard Arduino Nano ESP32 microcontroller, with the two central 22 kgf·cm ($\approx$ 2.16 N·m) servos synchronized via integrated gears mechanism. For inter-module coupling, each segment terminates in a triangular end equipped with five N42 neodymium magnets embedded inside protective covers: three situated at the vertices of the equilateral triangle, and two at the midpoints of its outward-facing edges. Because either end of a module can dock with any end of another module via these connecting edges, the hardware design imposes no directional pairing constraints.

Two target morphologies are assembled from this single module design: a three-module "trilobite" and a five-module "seal" (Fig. 2). Motion of an assembled morphology, whether as a stand-alone locomoting robot or as a parent robot performing module retrieval, is composed from three discrete motion primitives: a forward step, a left turn, and a right turn. These are executed by microcontroller firmware in response to commands issued over a serial link.

The module design relies on a magnetic docking interaction: an attractive force that falls off with the fourth power of separation distance, combined with velocity-dependent damping near contact. That study validated the numerical stability and reliable attachment behavior of the chosen docking parameters across a range of separation distances, independent of any specific target morphology.

## III. Teleoperated Collection and Assembly

Collection and assembly are currently carried out by direct human teleoperation rather than autonomous control. An operator observes the parent robot and the scattered spare modules directly, and one at a time issues a discrete motion command, a forward step, a left turn, or a right turn, through a browser-based control terminal. The terminal is connected over a USB serial link to a host microcontroller, which wirelessly broadcasts the corresponding command to the target modules; each module in turn sends a periodic heartbeat signal back to the host over the same wireless link to report its status. Guided by this real-time visual feedback, the operator steers the parent toward each target module in turn until its magnetic contact points engage. Docking is judged by the operator based on direct observation, without an independent physical



(a) Single module, folded 90°, close-up view (b) Trilobite, close-up view (c) Seal, close-up view

Fig. 2: (a) A single module folded to 90° between its two segments, illustrating the joint range described in Section II. (b) The assembled three-module trilobite morphology. (c) The assembled five-module seal morphology.

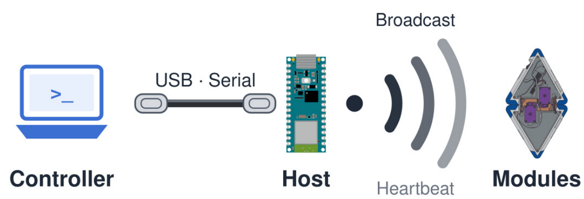



Fig. 3: Control system communication architecture. The operator issues commands from a browser-based control terminal on a laptop (Controller), connected over a USB serial link to a host microcontroller (Host). The Host wirelessly broadcasts each command to the modules, and each module sends a periodic heartbeat signal back to the Host over the same wireless link to report its status.

confirmation, such as force or current sensing, that a magnetic bond has actually formed.

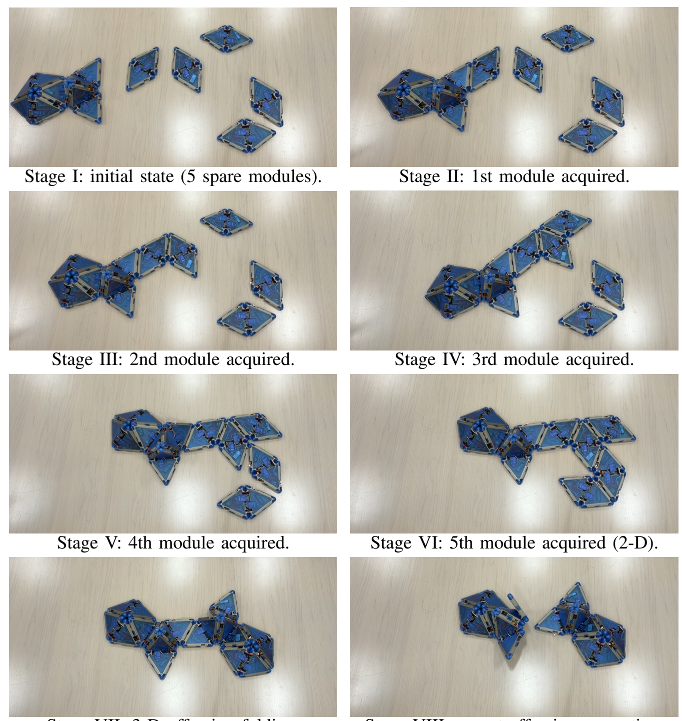
Once all modules for a given morphology have been collected, the operator triggers a pre-programmed separation sequence: distinct, timed motion paths for the parent and the newly formed offspring, executed as a single firmware-level command. The offspring is then free to move independently, and the operator issues several additional turn commands to it to demonstrate this independent locomotion.

Let $C$ denote the set of magnetic connections intended to open during separation. The desired topological result is

$$G_{2n} \setminus C = G_n^{(\mathrm{p})} \cup G_n^{(\mathrm{o})}, \qquad G_n^{(\mathrm{p})} \cap G_n^{(\mathrm{o})} = \varnothing. \tag{4}$$

Separation is achieved by antagonistic motions of the connected modules, which generate sufficient relative displacement and separating force to disengage the target magnetic interface while preserving the remaining connections. In the present implementation, satisfaction of this criterion is assessed from direct observation of the demonstration rather than by automated analysis of the connection graph.

For each trial, we record the number of successful dockings, the total collection time, whether separation succeeded, and whether the offspring could execute post-separation locomotion commands.



Stage I: initial state (5 spare modules). Stage II: 1st module acquired.
Stage III: 2nd module acquired. Stage IV: 3rd module acquired.
Stage V: 4th module acquired. Stage VI: 5th module acquired (2-D).
Stage VII: 3-D offspring folding. Stage VIII: parent-offspring separation.

Fig. 4: Eight stages of the five-module seal reproduction process, ordered from left to right and top to bottom. The parent sequentially acquires five initially loose environmental modules (I–VI). The acquired modules then fold from two-dimensional configurations into a three-dimensional offspring while still connected to the parent (VII). Selective magnetic separation produces two independent five-module seal robots (VIII): the persistent parent contains the five original modules, and the offspring contains the five acquired modules.

## IV. Preliminary Results

We demonstrate the full collection-and-assembly process on the five-module seal morphology: an assembled parent seal, under direct human teleoperation, located and magnetically docked with five scattered spare modules in sequence. The

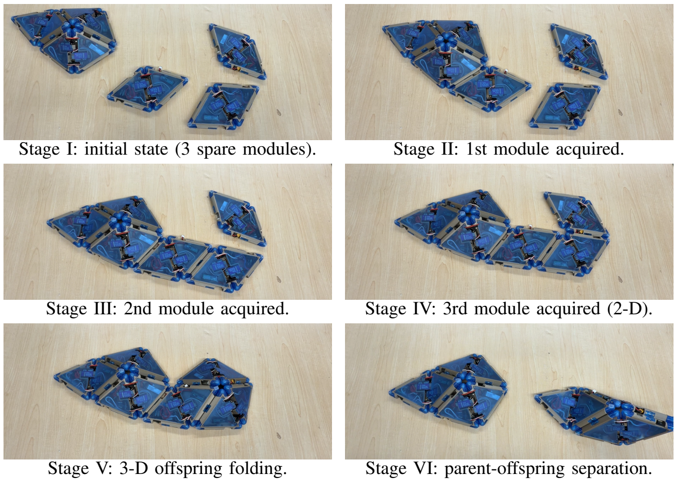



Fig. 5: Six stages of the three-module trilobite reproduction process, ordered from left to right and top to bottom. The parent sequentially acquires three initially loose environmental modules (I–IV). The acquired modules then fold from two-dimensional configurations into a three-dimensional offspring while still connected to the parent (V). Selective magnetic separation produces two independent three-module trilobite robots (VI): the persistent parent contains the three original modules, and the offspring contains the three acquired modules.

parent then executed the separation sequence, and a structurally identical offspring seal separated from the unchanged parent and was subsequently driven to execute several turns, demonstrating independent locomotion equivalent to that of the parent. Figure. 4 records eight stages of this complete reproduction process. In the initial state (I), the parent is a three-dimensional seal composed of five modules, and five additional modules are distributed separately in the environment. Sequential collection is shown in II–V: the parent has acquired one, two, three, and four modules, respectively, while the number of loose environmental modules decreases from four to one. After the fifth acquisition (VI), the composite system contains $2n = 10$ modules; all five acquired modules are connected to the parent but remain in their flat, two-dimensional configurations. The acquired modules subsequently fold into the three-dimensional seal morphology while the two five-module bodies remain connected (VII), producing the pre-separation configuration. Finally, the programmed antagonistic motion releases the target magnetic interface and produces a fully separated five-module parent and five-module offspring, both in their three-dimensional seal morphologies (VIII). The original five modules remain in the parent and the five acquired modules constitute the offspring, consistent with the provenance condition $\sigma = 1$. Because the two separated individuals share the same morphology, actuation, and docking hardware, each is physically equipped for subsequent module collection and separation, although recursive reproduction by the offspring has not yet been experimentally tested.

As shown in Figure. 5. The three-module trilobite morphology was assembled from the same module design, illustrating that a single module supports both target morphologies. The full collection-and-reproduction process was also demonstrated on the trilobite morphology: an assembled parent trilobite, under direct human teleoperation, located and magnetically docked with three scattered spare modules in sequence, executed the separation sequence, and produced a structurally identical offspring. The results reported here are single successful demonstrations on each morphology rather than statistically characterized trial series; a more complete quantitative evaluation is in preparation.

## V. Discussion and Future Work

Here, we show that a single module combining a magnetic docking interface with two actuated joints can support multiple target morphologies and physical, budding-style reproduction in an unstructured workspace [4]–[6]. Although module navigation and docking were teleoperated rather than autonomous, the demonstration confirms that the design mechanically supports the full end-to-end assembly and separation process.

The current implementation has several limitations that bound the scope of these results. Most significantly, collection and assembly are currently teleoperated rather than autonomous: navigation and docking decisions are made by a human operator based on direct observation, rather than by an onboard or vision-based controller. Docking success is judged by the operator from direct observation alone, without independent physical confirmation. The results reported above are single successful demonstrations on each morphology, rather than a statistically meaningful trial series. Finally, while the offspring produced is structurally identical to its parent and capable of independent locomotion, whether it can itself repeat the collection-and-assembly process to produce a further generation has not yet been tested.

Ongoing work targets these limitations directly: replacing teleoperation with autonomous, sensor-guided control of the collection-and-assembly process; repeating the demonstration across a larger number of independent trials on both morphologies; and testing whether an offspring can itself produce a subsequent generation. A more complete study incorporating these results is in preparation.

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